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Team ID	SWTID-2026-9440
Team Size	4
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Project Name	Machine Learning—Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Solution Requirements

Introduction

The Solution Requirements define the functional capabilities and expected behavior of the Machine Learning—Based Credit Card Approval Prediction System. These requirements specify the services ptov'ided by the application, user Interactions, system operations, data processing, and machine leaming functionality required to deliver accurate and efficient credit eard approval predictions.

1. Functional Requirements (FR)

	Requirement	Description	Priority
FR-OI	User Registration	Applicants shall legister securely using personal information.	High
FR-02	User I.Ã)gin	Registered users shall authenticate using valid credentials.	High
FR-03	Profile Management	Users shall update their personal information.	Medium
FR-04	Credit Card Application	Applicants shall submit financial and demographic information.	High
FR-05	Input Validation	The system shall validate mandatory' fields before submission.	High
FR-06	Data Pœpmcessing	The system shall clean and transform applicant data before prediction.	High
	Machine Learning Prediction	The system shall predict approval or rejection using the trained model.	High
FR-08	Prediction History	Store previous prediction results.	Medium
FR-09	Administrator Dashboard	Administrators shall monitor applications and paediction reports.	Medium
FR-10	Report Generation	Generate downloadable prediction reports and analytics.	Medium

Functional Requirements Summary

The functional requirements define the essential services required for secure user management, application prx»cessing, intelligent prediction, repolting, and system administration.

2. Non-Functional Requirements (NFR)

NFR ID	Requirement	Description	Acceptance Criteria
NFR-OI	Performance	Pœdiction shall be genemted quickly.	Response time less than 2 seconds.
NFR-02		Machine Learning model shall pmvide reliable pœdicuons.	Prediction accuracy yeater than 90%.
NFR-03	Security	Pli)tect user information using secure authentication and encryption.	HITPS, encrypted passwords, mle—based access.
NFR-04	Availability	Application shall remain continuously available.	99.9% uptime.
NFR-05	Scalability	Support increasing of users.	Minimum registered users.
NFR-06	Usability	Pmvide a simple and intuitive interface.	Easy navigation and responsive design.
NFR-07	Reliability	Ensure stable system operation.	Minimal prediction failures.
NFR-08	Maintainability	Support future enhancements and model updates.	Modular aœhitectuœ.

3. System Assumptions

- Users pmvide accurate applicant information.
- Internet connectivity is available.
- Tr,uned Machine Learning model is deployed successfully.
- Database server operates normally.
- Authorized users access administrative features.

4. System Constraints

- Prediction quality depends on dataset quality.
- Internet connection is required.
- Unsupported or Incomplete input may affect pœdiction.
- System performance depends on server resources.
- Future model retraining requiœs updated datasets.

Conclusion

The Solution Requirements establish the functional and non-functional expectations of the Machine Learning—Based Credit Card Approval Prediction System. These requirements ensure the application remains secure, scalable, reliable, user—friendly, and capable of delivering accurate credit card appmval pœdictions using Machine Learning techniques.